1. 1.8 V TO 3.3 V SYSTEM I2C CONTROL BUS

The control branch is isolated by its own PCA9306. Recheck total bus capacitance with the separate AUDIO-8X8 branch populated.

2. TWO 16-BIT GPIO EXPANDERS

U1001: 7-bit address 0x20; INT is test-point only and firmware polls status.

U1002: 7-bit address 0x21; INT is test-point only and firmware polls status.

TCA9535 outputs power up as inputs. External straps below define safe reset, mute, radio, and amplifier states.

W_DISABLE and WIFI_DISABLE are held active during boot; software releases them only after power and thermal checks pass.

3. EMC2305 FOUR-FAN CONTROL WITH FAILSAFE

ADDR_SEL 4.7k to 3.3 V selects 7-bit SMBus address 0x2E; internal clock is used.

CPU FAN

MODEM FAN

THA0412AD-TZW3 INTAKE

THA0412AD-TZW3 EXHAUST

Enclosure channels: 26.00 kHz base / divide 26 = 1.000 kHz PWM; direct duty with tach monitoring, 100% start and 30% minimum running duty.

Fan 1 is panel intake and fan 2 is panel exhaust. CPU and modem fans circulate locally beneath their dedicated mesh/keepout areas.

All four PWM nets pull high when U1020 is absent or unpowered, commanding the selected 4-wire fans to full speed.

6. INDEPENDENT WARM-WHITE PANEL LIGHTS

NIGHT LED 1

NIGHT LED 2

S1070 pin 5 is the rated low-side load output. Both 12 V LS102W courtesy lights switch together; pin 3 adds a dim green ON ring through R1070.

Hardware-only control: lights remain usable if the CM5 is booting, crashed, or powered down while AUX_12V is available.

4. THREE HIGH-ACCURACY TEMPERATURE SENSORS

TMP117 CM5 / 0x48

TMP117 MODEM / 0x49

TMP117 BOARD POWER / 0x4A

The three open-drain TMP117 alerts share TEMP_ALERT_N; firmware reads 0x48/0x49/0x4A to identify the source.

5. SIX TOP-PANEL STATUS LED DRIVERS

PWR GREEN

BACKUP VIOLET

WIFI BLUE

CELL AMBER

TEMP RED

AUDIO CYAN

TCA9535 sinks LED current. Final LED MPNs and resistor values follow top-panel brightness and color review.

U1000 PCA9306D
VREF1
VREF2
EN
SCL1
SDA1
SCL2
SDA2
GND
LOGIC_1V8
LOGIC_3V3
SYS_I2C7_SCL
SYS_I2C7_SDA
CTRL_I2C_SCL
CTRL_I2C_SDA
R1001 2.2k
R1002
R1003
R1004
R1005
PWR_MON_ALERT_N
U1001 TCA9535PWR_0x20
U1001_INT_N
CH_24V_N
CH_BAT_N
VALID_24V_N
VALID_BAT_N
BAT_LOW_N
FAN_ALERT_N
TEMP_ALERT_N
LED_PWR_N
LED_BACKUP_N
LED_WIFI_N
LED_CELL_N
LED_TEMP_N
LED_AUDIO_N
AUDIO_PRESENT_N
AUD_IRQ_N
U1002 TCA9535PWR_0x21
U1002_INT_N
AUDIO_ENABLE
AUD_ADC_RST_N
AUD_DAC_RST_N
AUD_DAC_MUTE_N
HEADSET_POWER_EN
HEADSET_AMP_EN
MODEM_FULL_CARD_POWER_OFF_N
MODEM_W_DISABLE1_N
MODEM_RESET_N
MODEM_SIM_MUX_SEL_OD
WIFI_DISABLE1_N
WAN2_RESET_LOCAL_N
LAN1_RESET_LOCAL_N
LAN2_RESET_LOCAL_N
MODEM_WAKE_N
MODEM_STATUS_LED_N
R1030
R1031
R1032
R1033
R1034
R1035
R1036
R1037
R1038
U1020 EMC2305-1-AP-TR
SMDATA
SMCLK
ADDR_SEL
CLK
ALERT
VDD
PWM1 TACH1 PWM2 TACH2 PWM3 TACH3 PWM4 TACH4 PWM5 TACH5
EMC2305_ADDR
CPU_FAN_PWM
CPU_FAN_TACH
MODEM_FAN_PWM
MODEM_FAN_TACH
INTAKE_FAN_PWM
INTAKE_FAN_TACH
EXHAUST_FAN_PWM
EXHAUST_FAN_TACH
R1020 4.7k 5%
R1021
C1020 100nF
CPU_FAN J1021
CPU_FAN_12V
MODEM_FAN J1022
MODEM_FAN_12V
THA0412AD-TZW3 INTAKE J1023
INTAKE_FAN_12V
THA0412AD-TZW3 EXHAUST J1024
EXHAUST_FAN_12V
F1021 2.0A hold
FAN_CPU_12V
F1022 1.0A hold
F1023 1.0A hold
F1024 1.0A hold
FAN_AUX_12V
CS7L2FR TOUCH LATCH S1070
CS_GREEN_RING_K
NIGHT_LIGHT_SINK
NIGHT_LIGHT_12V
R1070 ring limit
YIS LS102W 12V WARM WHITE D1070
YIS LS102W 12V WARM WHITE D1071
TMP117_CM5 U1050
TMP117_MODEM U1051
TMP117_BOARD_POWER U1052
SDA SCL ADD0 V+ ALERT GND
C1050 100nF
C1051 100nF
C1052 100nF
R1050
R1060
R1061
R1062
R1063
R1064
R1065
PWR_GREEN D1060
BACKUP_VIOLET D1061
WIFI_BLUE D1062
CELL_AMBER D1063
TEMP_RED D1064
AUDIO_CYAN D1065
PWR_GREEN_A
BACKUP_VIOLET_A
WIFI_BLUE_A
CELL_AMBER_A
TEMP_RED_A
AUDIO_CYAN_A
PWR_FLAG
FAN_CPU_12V
FAN_AUX_12V

TCA9535 0x20 captures source, fan, temperature, and power-monitor alerts
TMP117 sensors monitor CM5, modem, and board/power zones at 0x48/0x49/0x4A
PWM pull-ups provide full-speed hardware fallback if the controller or firmware fails
EMC2305 controls CPU, modem, intake, and exhaust fans independently
ProComm
Sheet: /
File: Thermal-IO.kicad_sch

Title: Radxa CM5 ProComm Carrier — Thermal and GPIO Control

Size: A0 | Date: 2026-08-14 | **Rev: A1**
KiCad E.D.A. 10.0.5-rc1 | Id: 1/1